STACKS, DISCRIMINANTS: LEMMA-GORENSTEIN-QUASI-FINITE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3507-3521 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 3523-3550, the complete author proof (26 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 21 of queue_discriminant.json (pb=1285, stmt 3507-3521, proof 3523-3550).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0C08: sources/stacks-discriminant/pinned-original-source.tex, source lines 3550–3550. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BV7: sources/stacks-discriminant/pinned-original-source.tex, source lines 3550–3550. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AX0: sources/stacks-discriminant/pinned-original-source.tex, source lines 3550–3550. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DW9: sources/stacks-discriminant/pinned-original-source.tex, source lines 3550–3550. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DWK: sources/stacks-discriminant/pinned-original-source.tex, source lines 2092–2103. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BUL: sources/stacks-discriminant/pinned-original-source.tex, source lines 3360–3375. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVG: sources/stacks-discriminant/pinned-original-source.tex, source lines 649–660. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : Y \to X$ be a quasi-finite morphism of Noetherian schemes. The following are equivalent*

(1) *f is Gorenstein,*
(2) *f is flat and the fibres of f are Gorenstein,*
(3) *f is flat and $\omega_{Y/X}$ is invertible (Remark 0BVG),*
(4) *for every $y \in Y$ there are affine opens $y \in V = \mathrm{Spec}(B) \subset Y$, $U = \mathrm{Spec}(A) \subset X$ with $f(V) \subset U$ such that $A \to B$ is flat and $\omega_{B/A}$ is an invertible B-module.*

Proof. Parts (1) and (2) are equivalent by definition. Parts (3) and (4) are equivalent by the construction of $\omega_{Y/X}$ in Remark 0BVG. Thus we have to show that (1)-(2) is equivalent to (3)-(4).

First proof. Working affine locally we can assume f is a separated morphism and apply Lemma 0BUL to see that $\omega_{Y/X}$ is the zeroth cohomology sheaf of $f^!\mathcal{O}_X$. Under both assumptions f is flat and quasi-finite, hence $f^!\mathcal{O}_X$ is isomorphic to $\omega_{Y/X}[0]$, see Duality for Schemes, Lemma 0BV7. Hence the equivalence follows from Duality for Schemes, Lemma 0C08.

Second proof. By Lemma 0DWK, we see that it suffices to prove the equivalence of (2) and (3) when X is the spectrum of a field k. Then $Y = \mathrm{Spec}(B)$ where B is a finite k-algebra. In this case $\omega_{B/A} = \omega_{B/k} = \mathrm{Hom}_k(B, k)$ placed in degree 0 is a dualizing complex for B, see Dualizing Complexes, Lemma 0AX0. Thus the equivalence follows from Dualizing Complexes, Lemma 0DW9. □